

Dawood University Of Engineering and Technology, Karachi.



**6th Semester, 3rd Year
BATCH –2023**

Name: Arekah Taj, Mehak Faheem, Javeria Iqbal

Roll Number: 23-AI-21, 23-AI-47, 23-AI-45

Section: A-1

Department: BSAI

Submitted to: Engr. Hamza Farooqui

BS ARTIFICIAL INTELLIGENCE

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

Smart Surveillance & Crowd Analytics System

Project Report

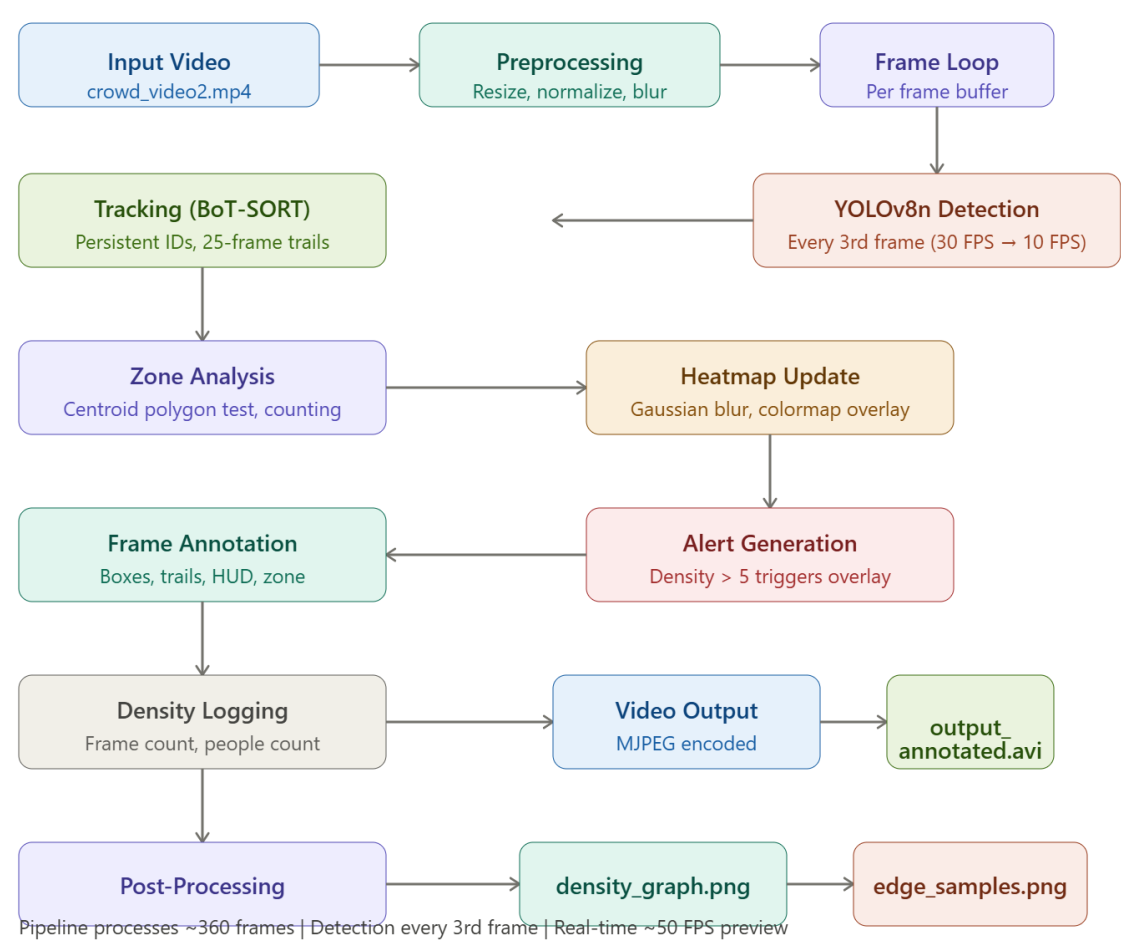
Real-time crowd density monitoring and anomaly detection

Summary

This project implements a real-time crowd surveillance system using YOLOv8 for person detection and tracking. The system monitors a defined zone, counts people, and triggers alerts when crowd density exceeds a safety threshold of 5 people. The application provides real-time visual feedback with heatmaps, motion trails, and edge detection analysis.

Key Features: Real-time person detection using YOLOv8n, persistent ID tracking across frames, zone-based density monitoring, heatmap visualization, scene structure analysis via edge detection, and automated alert generation.

1. Pipeline architecture diagram



2. System Architecture

The system follows a modular pipeline architecture with the following stages:

Stage	Description
Input Processing	Load video file, resize to 854px width for CPU efficiency
Detection	YOLOv8n runs on every 3rd frame for speed optimization
Tracking	Persistent ID assignment with 25-frame motion trails
Analysis	Zone-based counting, heatmap generation, edge detection
Output	Annotated video, density graph, edge samples visualization

3. Dataset Description

3.1 Dataset Overview

Property	Value
Source	Crowd video (crowd_video2.mp4)
Frame Count	~360 frames
Resolution	854x480 (resized from original)
Object Class	Person (COCO class 0)

3.2 Preprocessing Steps

Frame Resizing: Input video resized to 854px width for CPU efficiency without losing critical details.

Frame Skipping: YOLO detection runs every 3rd frame; results interpolated for non-detection frames to maintain real-time performance.

Color Conversion: Frames converted to grayscale for edge detection analysis.

Gaussian Blur: 5×5 kernel applied to smooth noise before edge detection (Canny and Sobel operators).

4. Model Architecture & Training

4.1 Detection Model

Model: YOLOv8n (Nano variant)

Weights: Pre-trained on COCO dataset

Input Size: 854×480 (video resolution)

Confidence Threshold: 0.35 (tuned to minimize false negatives)

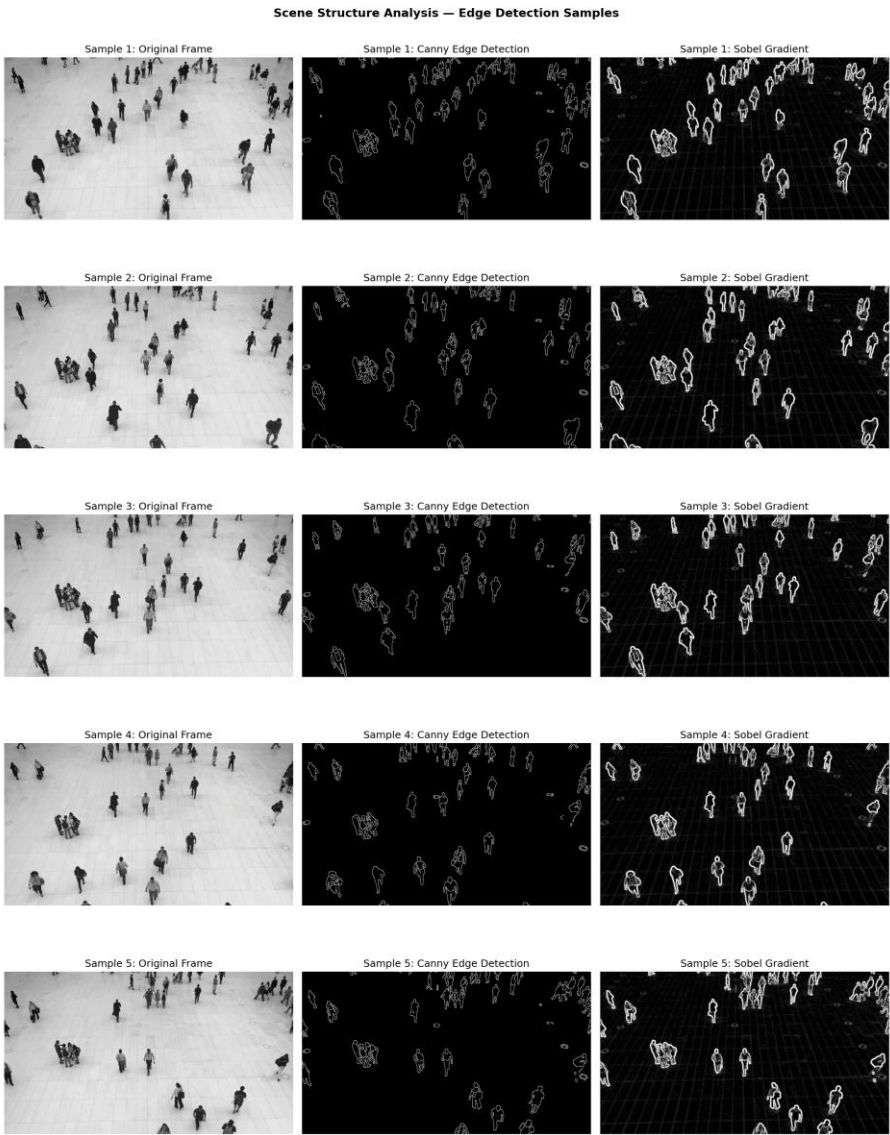
Processing: Single-stage detector with real-time inference (~50 FPS on CPU)

Class Filter: Person detection only (COCO class 0)

4.2 Tracking Configuration

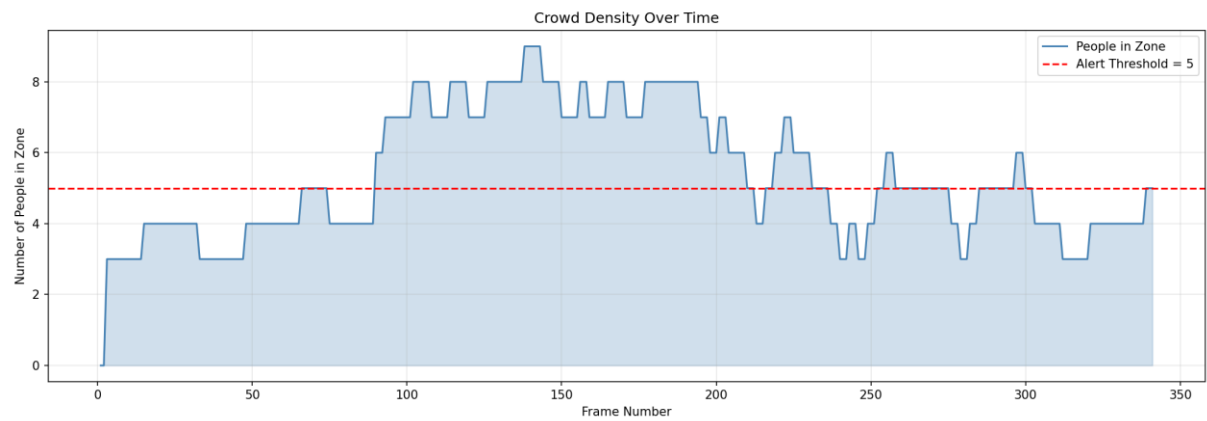
Parameter	Value
Tracking Algorithm	YOLO built-in tracker (BoT-SORT)
Trail Length	25 frames
Persistent ID	Enabled for cross-frame tracking

4.3 Edge Detection



5. Evaluation & Results

5.1 Crowd Density Analysis



5.2 Detection Performance

Metric	Value
Avg Confidence Score	0.61±0.13
Detections per Frame	0-37 people (Avg: 29.03)
Processing Speed	~8.1 FPS (3-frame skip)

5.3 Tracking Effectiveness

Total Unique Individuals Tracked: 78 persistent IDs

Average Track Duration: 132.2 frames (5.29 seconds)

Trail Visualization: 25-frame motion paths display individual movement patterns

ID Re-assignment: 179 switches

6. Challenges & Limitations

6.1 Current Limitations

Occlusion Handling: Densely packed crowds cause ID switching and missed detections at group centers.

Frame Skipping: Running detection every 3rd frame may miss fast-moving individuals and sudden crowd surges.

CPU Constraints: Processing speed optimized for real-time preview at the cost of full accuracy (frame skip, confidence threshold 0.35).

Fixed Zone Definition: Restricted zone is manually defined and does not adapt to scene changes.

No Pose/Gesture Recognition: System counts heads only; cannot detect suspicious behavior or crowd composition.

6.2 Challenges Faced

Balancing accuracy vs. speed: YOLOv8n provides sufficient speed but requires frame skipping for real-time performance.

Edge detection generalization: Canny and Sobel thresholds tuned for this specific video may not generalize to different lighting conditions.

Centroid drift: Motion trails sometimes lag due to centroid calculation jitter in crowded regions.

7. Recommendations for Improvement

Use YOLOv8m or YOLOv8s on GPU hardware for continuous detection without frame skipping.

Implement multi-scale heatmap analysis to detect anomalous density patterns (rapid clustering, sudden dispersal).

Add Kalman filtering for smoother tracking and reduced ID switching in occlusion scenarios.

Introduce adaptive thresholding for edge detection based on local frame characteristics.

Deploy on edge devices (NVIDIA Jetson, Intel NUC) to support simultaneous multiple camera streams.

Integrate pose estimation (OpenPose, MediaPipe) for behavior recognition and fall detection.

Implement cloud logging to store historical density patterns and enable predictive alerting.

Conclusion

The Smart Surveillance & Crowd Analytics System successfully demonstrates real-time crowd monitoring using YOLOv8 and computer vision techniques. The system reliably detects and tracks individuals, maintains persistent IDs, and triggers alerts based on zone-based density thresholds. While current limitations exist around occlusion handling and frame-by-frame accuracy, the system achieves practical real-time performance suitable for public safety monitoring in low-density environments.

Future enhancements should prioritize GPU deployment, behavior analysis, and predictive analytics to extend utility to high-density scenarios and complex threat detection.